

Auteur: Michael Zielinski
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$$\int_0^1 x^{\frac{1}{2}}(1-x)^{\frac{1}{2}} dx$$

$$x = \sin^2 u$$

$$dx = 2 \sin u \cos u \, du$$

$$x = 0 \Rightarrow u = 0$$

$$x = 1 \Rightarrow u = \frac{\pi}{2}$$

$$\int_0^1 x^{\frac{1}{2}}(1-x)^{\frac{1}{2}} dx = \int_0^{\frac{\pi}{2}} \sqrt{\sin^2 u} \sqrt{1 - \sin^2 u} \cdot 2 \sin u \cos u \, du$$

$$= \int_0^{\frac{\pi}{2}} \sin u \cos u \cdot 2 \sin u \cos u \, du$$

$$= \int_0^{\frac{\pi}{2}} 2 \sin^2 u \cos^2 u \, du$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin^2(2u) \, du$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{4} (1 - \cos(4u)) \, du$$

$$= \left[\frac{u}{4} - \frac{\sin(4u)}{16} \right]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{\pi}{8} - \frac{\sin(2\pi)}{16} \right) - \left(0 - \frac{\sin(0)}{16} \right)$$

$$= \frac{\pi}{8}$$

References